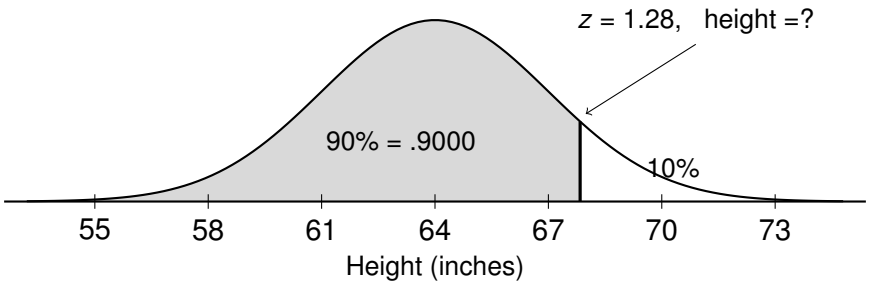
Normal Calculations in Reverse

Name _____

In 2.5, you knew the value and found the percent. Now you know the percent and find the value.

Example: Heights of adult women: mean 64 inches, SD 3 inches. How tall is a woman at the **90th percentile**?



Steps	**Example: 90th percentile**
1. Draw the curve. Shade the area to the **left**.	90% is to the left. The area is .9000.
2. Look **inside** the table for the area closest to it.	The closest area is .8997.
3. Read the z-score from the row and the column.	Row 1.2, column .08: $z = 1.28$
4. Find the value: value = mean + $z \times$ SD	$64 + 1.28 \times 3 = 64 + 3.84 = 67.84$

A woman at the 90th percentile is about **67.8 inches** tall.

"Top" and "bottom" questions

The **bottom 10%**	area to the left is .10
The **top 10%**	area to the left is $1 - .10 = .90$

Example: Below what height are the shortest 10% of women?

Area .1000. Closest in the table: .1003 at $z = -1.28$. $64 + (-1.28)(3) = 64 - 3.84 = 60.16$. About **60.2 inches**.

Gotcha: The table gives areas to the **left**. For "top" questions, subtract from 1 before you look in the table.

Use your z-table sheet for the problems.

Work with a partner. Compare answers when you finish each problem.

1. Use the z-table. Find the z-score at each percentile.

   (a) 80th percentile

   (b) 30th percentile

   (c) 97.5th percentile

   (d) 25th percentile

2. A test has scores with a normal distribution. The mean is 500 and the SD is 100.

   (a) What score is at the 90th percentile?

   (b) What score is at the 30th percentile?

3. A phone battery lasts a mean of 20 hours with an SD of 2 hours. The times have a normal distribution. The company replaces any battery in the bottom 10%. Below what time does a battery get replaced?

4. Apples have a mean weight of 150 grams and an SD of 10 grams. The weights have a normal distribution. The heaviest 15% are sold as “premium.” How heavy must an apple be to be premium?

5. Use the test from problem 2 (mean 500, SD 100). One question goes forward and one goes in reverse.

   (a) What percentile is a score of 640?

   (b) What score is at the 75th percentile?

6. **Challenge.** The heights of a type of plant have a normal distribution with a mean of 30 cm. 97.5% of the plants are shorter than 39.8 cm. Find the SD.